

APPROXIMATIONS TO THE DISTRIBUTIONS OF THE DETERMINANTS OF REAL AND COMPLEX MULTIVARIATE BETA MATRICES

Jack C. Lee

Wright State University

P.R. Krishnaiah

University of Pittsburgh

T.C. Chang

University of Cincinnati

Key words: Pearson type approximation, determinants of multivariate beta matrices, real and complex distributions.

SUMMARY

The distributions of certain powers of the determinants of the real and complex multivariate beta matrices are being approximated with Pearson's Type I distribution. The accuracy of this approximation is sufficient for practical purposes. Using this approximation, tables are presented for 5 per cent and 1 per cent upper points of the determinants of the real and complex multivariate beta matrices.

1. Introduction

The distribution of the determinant of the real multivariate beta matrix is useful in testing for the equality for the mean vectors of several multivariate normal populations. It is also useful in testing of the equality of the covariance matrices of two multivariate normal populations. Similarly the distribution of the determinant of the complex multivariate beta matrix plays an important role in testing for the equality of the spectral density matrices of two Gaussian, stationary, multiple time series since certain suitably defined estimates of the spectral density matrices of the above time series are approximately distributed as complex Wishart matrices.

The evaluation of the exact distribution function of the determinant of the real multivariate beta matrix was considered by Schatzoff (1966), Pillai and Gupta (1969), Mathai (1971) and Lee (1972) and some percentage points have been computed by these authors. Gupta (1971) constructed tables for the exact distribution of the determinant of the complex multivariate beta matrix in a few special cases. The exact distributions of the determinants of the real and complex multivariate beta matrices are complicated from a computational point of view.

In this paper, the authors have approximated the distributions of certain powers of

the determinants of the real and complex multivariate beta matrices with Pearson's Type I distribution. The accuracy of this approximation is sufficient for practical purposes. Using this approximation, the authors constructed tables for the distributions of the determinants of the real and complex multivariate beta matrices.

For a more complete set of tables, the reader is referred to a Technical Report by Lee, Krishnaiah and Chang (1975).

2. Distribution of the Determinant of Real Multivariate Beta Matrix

Let $S_1: p \times p$ and $S_2: p \times p$ be independently distributed as central Wishart matrices with n and q degrees of freedom respectively, and let $E(S_1/n) = E(S_2/q) = \Sigma$. Then $S_1(S_1 + S_2)^{-1}$ is known to be a multivariate beta matrix. Define $U_1 = |S_1| / |S_1 + S_2|$. The moments of U_1 are known (see Wilks (1932)) to be

$$(2.1) \quad E(U_1^h) = \prod_{i=1}^p \left\{ \frac{\Gamma[\frac{1}{2}(n+1-i)+h] \Gamma[\frac{1}{2}(n+q+1-i)]}{\Gamma[\frac{1}{2}(n+1-i)] \Gamma[\frac{1}{2}(n+q+1-i)+h]} \right\}$$

where $\Gamma(\cdot)$ is the complete gamma function. Using the first four moments of $U_1^{1/b}$, we approximated the distribution of $U_1^{1/b}$ with Pearson's Type I distribution, where b is a suitably chosen integer. For the choice of b in $U_1^{1/b}$, we have essentially tried to enlarge the magnitude of U so that the computational accuracy is retained. Of course, we also chose b so that the type I distribution was selected. For $M \leq 5$, we took $b=8$, for $M \geq 30$, $b=2$ and for the rest $b=4$. Using this approximation, the values of c_1 are computed for $\alpha = 0.05$ and 0.01 , $M = n - p + 1 = 1(1) 20, 24, 30, 60, 120$ and certain values of (p, q) where

$$(2.2) \quad P[C_1 \leq c_1] = 1 - \alpha,$$

$C_1 = -\{n - \frac{1}{2}(p - q + 1)\} \log U_1 / \chi_{pq}^2(\alpha)$ and $\chi_{pq}^2(\alpha)$ is the upper α -point of

the χ^2 distribution with pq degrees of freedom. The critical values c_1 are given in Table 2. The reader is referred to Lee, Krishnaiah and Chang (1975) for the values of c_1 for $\alpha = 0.10; 0.025; 0.005; 0.90; 0.95; 0.99; 0.995$. The percentage points given in Table 2 complement the values given in the literature (see Schatzoff (1966), Pillai and Gupta (1969), Mathai (1971) and Lee (1972)) which were computed by using complicated exact expressions.

To check for the accuracy of the tables in this paper, the authors compared some of the values obtained by Pearson type approximation with the corresponding exact values. These comparisons are given in Table 1. In this table, α is given by $P[C_1 \leq c_1] = 1 - \alpha$.

This table indicates that the accuracy of Pearson type approximation is sufficient for practical purposes. The asymptotic expression derived by Box (1949) was also used to check for the accuracy of the tables in this paper. The coefficients of this expression up to order N^{-13} were given in Lee, Krishnaiah and Chang (1976).

TABLE 1
COMPARISON OF THE PEARSON TYPE APPROXIMATION
WITH EXACT EXPRESSION OF THE DISTRIBUTION OF C_1

($\alpha = 0,05$)

M	p = 3 q = 3		p = 5 q = 7	
	c_1	Exact	c_1	Exact
1	1,354	1,359	1,530	1,535
5	1,034	1,035	1,090	1,090
10	1,010	1,011	1,035	1,035
16	1,005	1,005	1,017	1,017
20	1,003	1,003	1,012	1,012
40	1,001	1,001	1,003	1,003

TABLE 2*
Upper Percentage Points of the Distribution of C_1

M	p = 9 , q = 11		p = 9 , q = 13	
	0,050 α	0,010	0,050 α	0,010
1	1,542	1,650	1,554	1,659
2	1,311	1,348	1,326	1,363
3	1,219	1,240	1,234	1,255
4	1,168	1,182	1,182	1,195
5	1,134	1,144	1,147	1,157
6	1,111	1,118	1,122	1,130
7	1,093	1,100	1,104	1,110
8	1,080	1,085	1,090	1,095
9	1,070	1,074	1,078	1,082
10	1,061	1,065	1,069	1,073
12	1,048	1,051	1,055	1,058
14	1,039	1,042	1,045	1,047
16	1,033	1,035	1,038	1,040
18	1,028	1,029	1,032	1,034
20	1,024	1,025	1,028	1,029
24	1,018	1,019	1,021	1,022
30	1,013	1,013	1,015	1,016
40	1,008	1,008	1,009	1,010
60	1,004	1,004	1,005	1,005
120	1,001	1,001	1,001	1,001
∞	1,000	1,000	1,000	1,000
$\chi^2_{pq}(\alpha)$	123,2250	134,6420	143,2460	155,4960

*The entries in this table are the values of c_1 where

$P[C_1 \leq c_1] = (1 - \alpha)$, $C_1 = -\{n - \frac{1}{2}(p - q + 1)\} \log U_1 / \chi^2_{pq}(\alpha)$, $M = n - p + 1$,
and $\chi^2_{pq}(\alpha)$ is the upper α -point of chi-square with pq degrees of freedom.

TABLE 2 (Continued)

M	p = 9 , q = 15		p = 9 , q = 17	
	α		α	
	0,050	0,010	0,050	0,010
1	1,568	1,673	1,585	1,690
2	1,343	1,380	1,360	1,398
3	1,250	1,271	1,266	1,287
4	1,196	1,210	1,210	1,225
5	1,160	1,170	1,173	1,184
6	1,134	1,142	1,146	1,154
7	1,115	1,121	1,126	1,132
8	1,099	1,105	1,110	1,115
9	1,087	1,092	1,096	1,102
10	1,077	1,081	1,086	1,090
12	1,062	1,065	1,070	1,073
14	1,051	1,054	1,058	1,060
16	1,043	1,045	1,049	1,051
18	1,037	1,038	1,042	1,043
20	1,032	1,033	1,036	1,038
24	1,024	1,026	1,028	1,029
30	1,017	1,018	1,020	1,021
40	1,011	1,012	1,013	1,013
60	1,006	1,006	1,007	1,007
120	1,002	1,002	1,002	1,002
∞	1,000	1,000	1,000	1,000
$\chi^2_{pq}(\alpha)$	163,1160	176,1380	182,8650	196,6090

TABLE 2 (Continued)

M	p = 11 , q = 11		p = 11 , q = 13	
	α		α	
	0,050	0,010	0,050	0,010
1	1,536	1,638	1,538	1,637
2	1,316	1,351	1,324	1,359
3	1,226	1,246	1,236	1,256
4	1,175	1,189	1,185	1,198
5	1,142	1,152	1,151	1,161
6	1,118	1,125	1,127	1,134
7	1,100	1,106	1,108	1,114
8	1,086	1,092	1,094	1,099
9	1,075	1,080	1,082	1,087
10	1,067	1,070	1,073	1,077
12	1,053	1,056	1,059	1,061
14	1,044	1,046	1,048	1,051
16	1,036	1,038	1,041	1,042
20	1,027	1,028	1,030	1,031
24	1,020	1,021	1,023	1,024
30	1,014	1,015	1,017	1,017
40	1,009	1,010	1,011	1,011
60	1,005	1,005	1,005	1,005
120	1,001	1,001	1,002	1,002
∞	1,000	1,000	1,000	1,000
$\chi^2_{pq}(\alpha)$	147,6740	160,1000	171,9070	185,2560

3. Distribution of the Determinant of Complex Multivariate Beta Matrix

Let $A_1: p \times p$ and $A_2: p \times p$ be independently distributed as central complex Wishart matrices with n and q degrees of freedom and let $E(A_1/n) = E(A_2/q) = \Sigma$. Then $A_1(A_1 + A_2)^{-1}$ is known to be a complex multivariate beta matrix. Now let $U_2 = |A_1(A_1 + A_2)^{-1}|$. The h -th moment of U_2 is given by

$$(3.1) \quad E(U_2^h) = \prod_{j=1}^p \left\{ \frac{\Gamma(n+h-j+1) \Gamma(n+q-j+1)}{\Gamma(n-j+1) \Gamma(n+h+q-j+1)} \right\}.$$

Using the first four moments of $U_2^{1/b}$, we approximate the distribution of $U_2^{1/b}$ with Pearson's Type I distribution where b is a properly chosen integer. For $M \geq 20$ we chose $b = 1$ and for $M < 20$, $b = 2$. By making use of this approximation, values of c_2 are computed, where

$$(3.2) \quad P[C_2 \leq c_2] = 1 - \alpha,$$

$C_2 = -(2n + q - p) \log U_2 / \chi^2_{2pq}(\alpha)$ and $\chi^2_{2pq}(\alpha)$ is the upper α -point of χ^2 with $2pq$ degrees of freedom. The values of c_2 are computed for $\alpha = 0,05$ and $0,01$; $M = 1(1)10(2)20; 30; 60; 120$ where $M = n - p + 1$. These values are given in Table 4. For the values of c_2 for $\alpha = 0,10; 0,025; 0,005; 0,90; 0,95; 0,99; 0,995$ the reader is referred to Lee, Krishnaiah and Chang (1975).

To check for the accuracy of the entries in Table 4, we compared some of the values obtained by Pearson type approximation with the corresponding exact values. These comparisons are given in Table 3.

In Table 3, α is defined by (3.2). Also, the values under the column "L-K-C" are the values of c_2 obtained by the authors using Pearson type approximation whereas the values under the column "Exact" are the corresponding values given by Gupta (1971). Table 3 indicates that the accuracy of Pearson type approximation is sufficient for practical purposes.

TABLE 3

Comparison of the Pearson Type Approximation with Exact Expression for the Distribution of C_2

		p = 2 , q = 3		p = 2 , q = 20	
M	α	L-K-C	Exact	L-K-C	Exact
1	0,05	1,286	1,289	1,928	1,932
	0,01	1,350	1,349	2,085	2,080
5	0,05	1,029	1,029	1,243	1,243
	0,01	1,033	1,033	1,262	1,262
9	0,05	1,010	1,011	1,128	1,128
	0,01	1,012	1,012	1,137	1,137

TABLE 4*
UPPER PERCENTAGE POINTS* OF THE DISTRIBUTION OF C_2

M	p = 3 , q = 3		p = 3 , q = 7	
	α		α	
	0,050	0,010	0,050	0,010
1	1,310	1,378	1,446	1,528
2	1,132	1,153	1,225	1,253
3	1,076	1,087	1,145	1,160
4	1,050	1,057	1,103	1,113
5	1,036	1,041	1,078	1,085
6	1,027	1,030	1,061	1,066
7	1,021	1,024	1,049	1,054
8	1,017	1,019	1,041	1,044
9	1,014	1,016	1,034	1,037
10	1,011	1,013	1,029	1,032
12	1,008	1,009	1,022	1,024
14	1,007	1,007	1,017	1,019
16	1,005	1,006	1,014	1,015
18	1,004	1,005	1,011	1,012
20	1,003	1,004	1,010	1,010
30	1,002	1,002	1,005	1,005
60	1,000	1,000	1,001	1,001
120	1,000	1,000	1,000	1,000
$\chi^2_{2pq}(\alpha)$	28,8693	34,8053	58,1240	66,2062

* The entries of this table are the values of c_2 where

$$P[C_2 \leq c_2] = 1 - \alpha, C_2 = - \{ (2n + q - p) \log U_2 \} / \chi^2_{2pq}(\alpha).$$

TABLE 4 (Continued)

M	p = 3 , q = 10		p = 4 , q = 4	
	α		α	
	0,050	0,010	0,050	0,010
1	1,538	1,630	1,354	1,425
2	1,293	1,326	1,166	1,189
3	1,197	1,216	1,102	1,114
4	1,145	1,158	1,070	1,078
5	1,113	1,122	1,052	1,057
6	1,090	1,097	1,040	1,044
7	1,074	1,080	1,032	1,035
8	1,062	1,067	1,026	1,028
9	1,053	1,057	1,021	1,024
10	1,046	1,049	1,018	1,020
12	1,035	1,038	1,013	1,015
14	1,028	1,030	1,010	1,011
16	1,023	1,025	1,008	1,009
18	1,019	1,020	1,007	1,007
20	1,016	1,017	1,006	1,006
30	1,008	1,009	1,003	1,003
60	1,002	1,002	1,001	1,001
120	1,001	1,001	1,000	1,000
$\chi^2_{2pq}(\alpha)$	79,0819	88,3794	46,1943	53,4858

TABLE 4 (Continued)

M	p = 4 , q = 7		p = 4 , q = 10	
	α		α	
	0,050	0,010	0,050	0,010
1	1,416	1,490	1,485	1,562
2	1,217	1,242	1,272	1,299
3	1,142	1,156	1,186	1,202
4	1,103	1,112	1,139	1,149
5	1,078	1,085	1,108	1,116
6	1,062	1,067	1,088	1,094
7	1,051	1,054	1,073	1,077
8	1,042	1,045	1,061	1,065
9	1,035	1,038	1,053	1,056
10	1,030	1,033	1,046	1,048
12	1,023	1,025	1,035	1,037
14	1,018	1,020	1,028	1,030
16	1,015	1,016	1,023	1,024
18	1,012	1,013	1,019	1,020
20	1,010	1,011	1,016	1,017
30	1,005	1,005	1,008	1,009
60	1,001	1,001	1,002	1,003
120	1,000	1,000	1,001	1,001
$\chi^2_{2pq}(\alpha)$	74,4683	83,5134	101,8790	112,3290

TABLE 4 (Continued)

M	p = 5 , q = 5		p = 5 , q = 8	
	α		α	
	0,050	0,010	0,050	0,010
1	1,378	1,448	1,419	1,488
2	1,191	1,214	1,230	1,254
3	1,123	1,135	1,156	1,169
4	1,087	1,095	1,115	1,124
5	1,066	1,072	1,089	1,096
6	1,052	1,056	1,072	1,077
7	1,042	1,045	1,059	1,063
8	1,035	1,037	1,050	1,053
9	1,029	1,031	1,042	1,045
10	1,025	1,027	1,037	1,039
12	1,019	1,020	1,028	1,030
14	1,015	1,016	1,022	1,024
16	1,012	1,013	1,018	1,019
18	1,010	1,011	1,015	1,016
20	1,008	1,009	1,013	1,013
30	1,004	1,004	1,007	1,007
60	1,001	1,001	1,002	1,002
120	1,000	1,000	1,001	1,001
$\chi^2_{2pq}(\alpha)$	67,5048	76,1539	101,8790	112,3290

TABLE 4 (Continued)

M	p = 5 , q = 10		p = 6 , q = 6	
	α		α	
	0,050	0,010	0,050	0,010
1	1,453	1,522	1,392	1,459
2	1,259	1,284	1,210	1,233
3	1,180	1,194	1,140	1,152
4	1,136	1,145	1,102	1,110
5	1,107	1,114	1,079	1,084
6	1,087	1,092	1,063	1,067
7	1,072	1,077	1,051	1,055
8	1,062	1,065	1,043	1,046
9	1,053	1,056	1,037	1,039
10	1,046	1,049	1,031	1,034
12	1,036	1,038	1,024	1,026
14	1,029	1,030	1,019	1,020
16	1,024	1,025	1,015	1,017
18	1,020	1,021	1,013	1,014
20	1,017	1,017	1,011	1,011
30	1,009	1,009	1,006	1,006
60	1,003	1,003	1,002	1,002
120	1,001	1,001	1,000	1,000
$\chi^2_{2pq}(\alpha)$	124,3420	135,8070	92,8083	102,8160

TABLE 4 (Continued)

M	p = 6 , q = 10		p = 7 , q = 7	
	α		α	
	0,050	0,010	0,050	0,010
1	1,433	1,497	1,401	1,464
2	1,253	1,276	1,225	1,247
3	1,178	1,191	1,154	1,167
4	1,135	1,144	1,115	1,124
5	1,107	1,114	1,090	1,096
6	1,088	1,093	1,073	1,077
7	1,074	1,078	1,060	1,064
8	1,063	1,066	1,051	1,054
9	1,054	1,057	1,044	1,046
10	1,047	1,050	1,038	1,040
12	1,037	1,039	1,029	1,031
14	1,030	1,031	1,023	1,025
16	1,025	1,026	1,019	1,020
18	1,021	1,022	1,016	1,017
20	1,018	1,018	1,014	1,014
30	1,009	1,010	1,007	1,007
60	1,003	1,003	1,002	1,002
120	1,001	1,001	1,001	1,001
$\chi^2_{2pq}(\alpha)$	146,5670	158,9500	122,1080	133,4760

ACKNOWLEDGEMENT

The authors are grateful to the referee for some helpful comments.

REFERENCES

- ANDERSON, T.W. (1958). *An introduction to multivariate statistical analysis*. John Wiley & Sons, Inc., New York.
- BOX, G.E.P. (1949). A general distribution theory for a class of likelihood criteria. *Biometrika* **36** 317-346.
- GOODMAN, N.R. (1963). Statistical analysis based on a certain multivariate complex Gaussian distribution (an introduction). *Ann. Math. Statist.* **34** 152-176.
- GUPTA, A.K. (1971). Distribution of Wilks' likelihood-ratio criterion in the complex case. *Ann. Inst. Statist. Math.* **23** 77-87.
- LEE, J.C., KRISHNAIAH, P.R. and CHANG, T.C. (1975). Approximations to the distributions of the determinants of real and complex multivariate beta matrices. ARL TR 75-0168.
- LEE, J.C., KRISHNAIAH, P.R. and CHANG, T.C. (1976). On the distribution of the likelihood ratio test statistic for compound symmetry. *South African Statist. J.* **10** 49-62.
- LEE, Y.S. (1972). Some results on the distribution of Wilks' likelihood-ratio criterion. *Biometrika* **59** 649-664.
- MATHAI, A.M. (1971). On the distribution of the likelihood-ratio criterion for testing linear hypotheses on regression coefficients. *Ann. Inst. Statist. Math.* **23** 181-197.
- PILLAI, K.C.S. and GUPTA, A.K. (1969). On the exact distribution of Wilks' criterion. *Biometrika* **56** 109-118.
- SCHATZOFF, M. (1966). Exact distribution of Wilks' likelihood-ratio criterion. *Biometrika* **53** 347-358.
- WILKS, S.S. (1932). Certain generalizations in the analysis of variance. *Biometrika*. **24** 471-494.

Manuscript received 30 September 1976, revised 17 January 1977.